Cor Jesu College

Engineering Data Analysis

Study Habits and Travel to Campus

TAKE-HOME APPLIED EXAM | CHAPTERS 1–3

SUBMITTED BY

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Study habits and travel to campus

COR JESU COLLEGE | DESCRIPTIVE STATISTICS REPORT

1 Decision question

What do study-hour patterns and one-way commute times suggest for planning optional study-support sessions?

2 Sample, method, and measurement

Population: Currently enrolled undergraduate students at Cor Jesu College in the Freshman, Sophomore, Junior, and Senior categories.
Sample: 32 valid respondents from 32 submitted records; 0 excluded by validation. Collection: 2026-10-14 to 2026-10-14. Sampling method: convenience. An examination day was deliberately chosen to reach students attending campus. Available and willing students in the lobby of Norbert Building were approached in person.
A statistic describes these records (for example, their mean); a parameter describes the whole target population. These records do not establish the population mean.
Q1 is ordinal categorical. Q2 records discrete study hours. Q3 is continuous time recorded in whole minutes. Hours and minutes use ratio scales; the survey codes walking/on-campus travel as 0.
Rounding: Whole-number hours and minutes were copied from the response sheet without further rounding; no 8+ answer occurs.
Grouping: Q2 class width 2 h; midpoints 0.5, 2.5, 4.5, 6.5, 8.5. Q3 width 10 min; whole-minute limits have boundaries +/-0.5 min. Midpoint = (lower + upper boundary) / 2.

3 Frequency distributions

Q1: Year level

Category	f	Rel. %	Cum. f
Freshman	5	15.6%	5
Sophomore	16	50.0%	21
Junior	4	12.5%	25
Senior	7	21.9%	32

Q2: Study hours

Hours / class	f	Rel. %	Cum. f
0–1	11	34.4%	11
2–3	13	40.6%	24
4–5	8	25.0%	32
6–7	0	0.0%	32
8–9	0	0.0%	32

Q3: One-way commute time (minutes)

Minute class	Frequency	Relative %	Cumulative f
0–9	2	6.2%	2
10–19	5	15.6%	7
20–29	14	43.8%	21
30–39	5	15.6%	26
40–49	5	15.6%	31
50–59	1	3.1%	32

All three tables use n = 32; relative frequency = f / n. Percentages may differ from 100% by rounding. Cumulative year-level counts follow form order only.

Description of the recorded values

4 Descriptive statistics and first two figures

Statistics below use ungrouped valid records. Figures 1 and 2 show year-level composition and recorded study hours. Most common year level(s): Sophomore (16 each). Missing year-level categories: none. Sample variance and standard deviation use n - 1. Inclusive linear quantiles match Sheets INC; Bluman section 3-3 hand rules may differ.

Statistic	Study hours (h)	Commute time (min)
Mean	2.34	25.16
Median	2.00	25.00
Mode(s)	1, 2	20, 25
Minimum	0.00	0.00
Maximum	5.00	55.00
Range	5.00	55.00
Sample variance (squared units)	2.49	140.07
Sample standard deviation	1.58	11.84
First quartile (Q1)	1.00	20.00
Third quartile (Q3)	3.25	30.00
Interquartile range (IQR)	2.25	10.00
10th percentile (P10)	0.10	12.30
90th percentile (P90)	4.90	40.00
Lower outlier fence	-2.38	5.00
Upper outlier fence	6.62	45.00

All study-hour answers are finite whole-hour values; no open-ended answer occurs. The 1.5 x IQR rule flags 0 study-hour and 2 commute record(s). Commute flags: ID 24 = 55 min; ID 30 = 0 min. Flagged values are retained for review. The 0-minute value is allowed by the survey rule and is retained.

Figure 1. Year-level frequencies

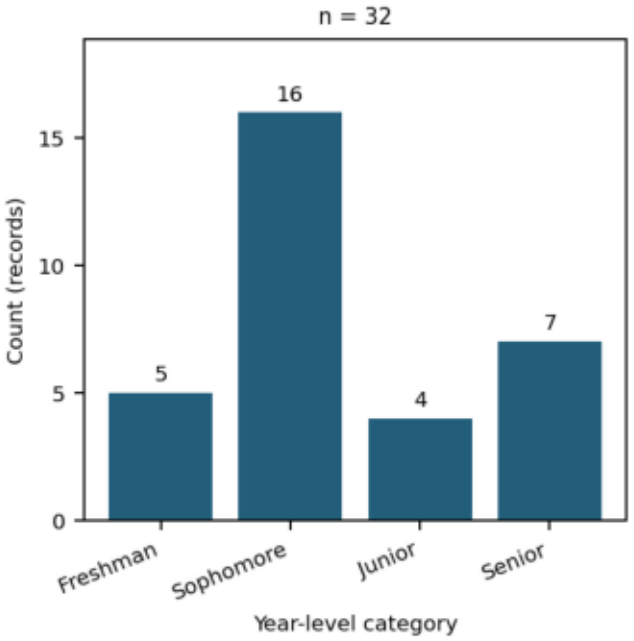
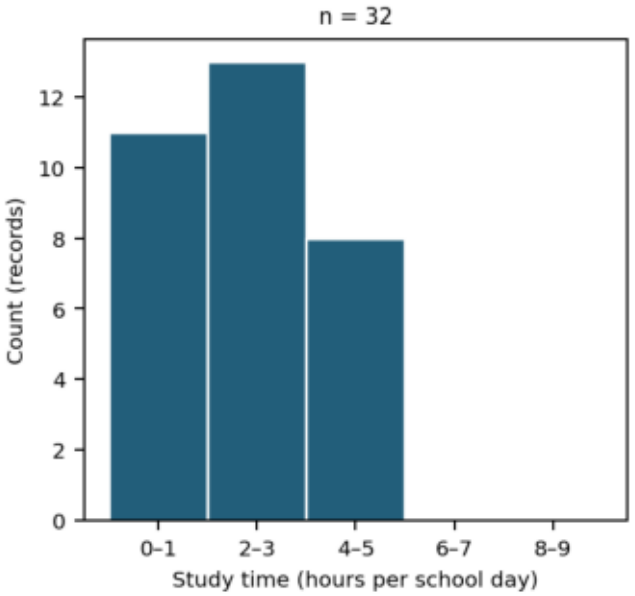


Figure 2. Recorded daily study hours



Patterns, decision, and limitations

Figures 3 and 4 show study-hour frequencies and cumulative counts; Figures 5 and 6 show commute shape and IQR outliers. Study hours: The recorded distribution has no strong moment skew (moment skewness 0.217; descriptive threshold ±0.5). Commute: The recorded distribution has no strong moment skew (moment skewness 0.382; descriptive threshold ±0.5). These are not normality tests. The polygon closes at zero; the ogive starts at zero below the first class.

Figure 3. Study-hour frequency polygon

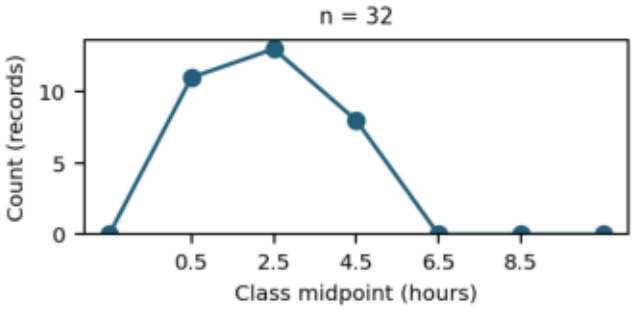


Figure 4. Study-hour cumulative frequencies

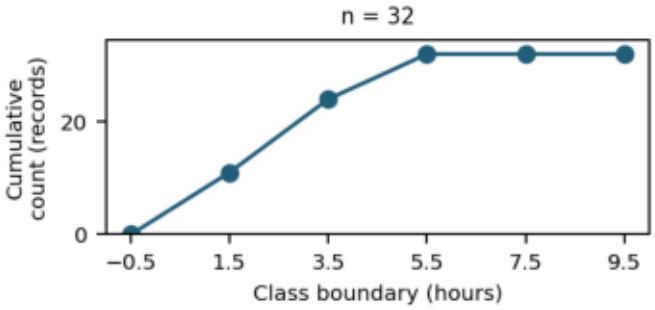


Figure 5. Recorded one-way commute times

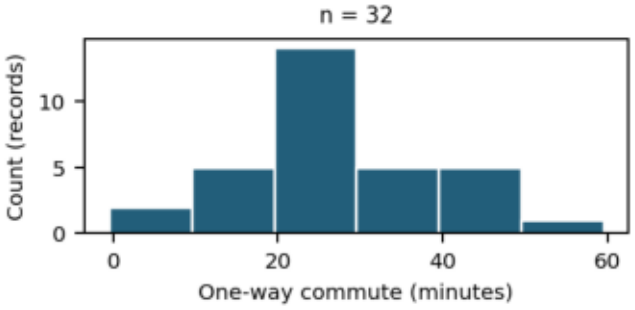
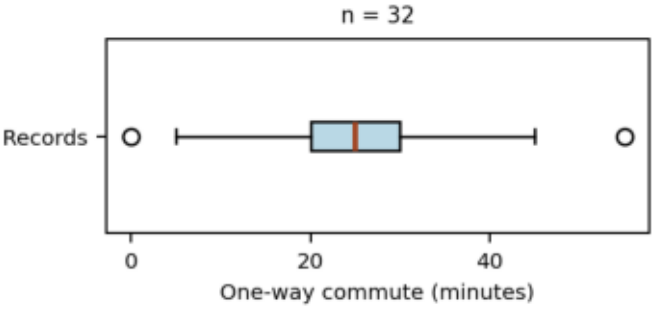


Figure 6. Commute distribution and outliers



5 Recommendation

The sample reports a median of 2 study hours/day and a 25-minute median one-way commute (P90: 40 minutes). Consider study-support sessions alongside existing classes, with a remote option to reduce additional travel. Confirm preferred times before setting a schedule; these summaries do not establish session demand.

6 Limitations and next evidence

Most vulnerable assumption: Convenience sampling at one building may not represent all enrolled undergraduates; sophomores make up half the sample. The exam calls for probability sampling. Next evidence: Repeat collection with a probability sample across year levels and campus locations, and ask about preferred study-session times. Self-report and the zero-for-walking rule limit interpretation. Descriptive associations do not establish cause and effect.

References: Allan G. Bluman (2012), Elementary Statistics: A Step by Step Approach, 8th ed., McGraw-Hill; sections 1-1, 1-2, 1-3; 2-1, 2-2, 2-3; 3-1, 3-2, 3-3, 3-4. Cor Jesu College applied exam and supplied survey form.